Exercise 1. Let S be an ordered set. Let $A \subset S$ be a nonempty finite subset. Then A is bounded. *Hint: Use induction.*

Proof. Let S be any ordered set. Assume $A \subset S$ and $A \neq \emptyset$ and $|A| = n$. We want to prove that A is bounded. Since $|A| = n$, $\exists$ bijection $f : A \to \{1, ..., n\}$. Then $f(A) \subset \mathbb{N}$ and $f(A) \neq \emptyset$. By the Well-Ordering Property of $\mathbb{N}$, $\exists$ least element x_1 of A. Suppose $x_1 < \cdots < x_i$ is a partial ordering of A. Then $f(A \setminus \bigcup_{k=1}^{i}\{x_k\}) \subset \mathbb{N}$ and $f(A \setminus \bigcup_{k=1}^{i}\{x_k\}) \neq \emptyset$. By the Well-Ordering Property of $\mathbb{N}$, $\exists$ least element x_{i+1} of $A \setminus \bigcup_{k=1}^{i}\{x_k\}$. Since $|A| = n$, $f(A \setminus \bigcup_{k=1}^{n}\{x_k\}) = \emptyset$. By the Principle of Induction, $x_1 < \cdots < x_n$ is a complete ordering of A. Thus, $\forall x \in A$ $(x_1 \leq x \leq x_n)$, as needed. □

Exercise 2. Let F be an ordered field and $x, y, z \in F$. Prove if $x < 0$ and $y < z$, then $xy > xz$.

Proof. If $x < 0$ and $y < z$, then

$$xy = x(y - z) + xz = (-x)(z - y) + xz > 0 + xz = xz,$$

as needed. □

Exercise 3. Let S be an ordered set. Let $A \subset S$ and suppose b is an upper bound for A. Suppose $b \in A$. Show that $b = \sup A$.

Proof. Let S be any ordered set. Suppose $A \subset S$ and b is an upper bound for A and $b \in A$. If $b' \in S$ and $b' < b$, then $b' < b \in A$ and so b' is not an upper bound for A. Thus, $b = \sup A$, as needed. □

Exercise 4. Let S be an ordered set. Let $A \subset S$ be a nonempty subset that is bounded above. Suppose $\sup A$ exists and $\sup A \notin A$. Show that A contains a countably infinite subset.

Proof. Let S be any ordered set. Assume $A \subset S$ is nonempty and bounded above. Suppose $\sup A \in S \setminus A$. Then $\forall x \in A$ $\exists y \in A : x < y < \sup A$; otherwise, $\sup A = x \in A$ xor $\sup A = y \in A$, a contradiction. We want to construct a sequence $\{x_n\}_{n=1}^{\infty}$ of distinct entries in A. Choose $x_1 \in A$. Suppose $x_1 < \cdots < x_n \in A$. Then $\exists x_{n+1} \in A : x_n < x_{n+1} < \sup A$; else, $\sup A = x_n \in A$ xor $\sup A = x_{n+1} \in A$, a contradiction. By induction, $\{x_n\}_{n=1}^{\infty}$ is a monotone increasing sequence of distinct entries in A. We want to show that A contains a countably infinite subset E. Define the subset $E := \{x_n \in A : \forall n \in \mathbb{N}\}$ of S. Then $E = \{x_n \in A : \forall n \in \mathbb{N}\} \subset A$ and $|E| = |\{x_n \in A : \forall n \in \mathbb{N}\}| = |\mathbb{N}|$. Thus, A contains a countably infinite subset E, as needed. □

Exercise 5. Prove the arithmetic-geometric mean inequality. That is, for two positive real numbers x, y, we have

$$\sqrt{xy} \leq \frac{x + y}{2}.$$

Furthermore, equality occurs if and only if $x = y$.

Proof. Let $x, y \in \mathbb{R}$ and $x, y > 0$. We want to prove that

$$\sqrt{xy} \leq \frac{x + y}{2}.$$

Thus, we have

$$(x - y)^2 \geq 0 \implies 4xy \leq x^2 + 2xy + y^2 \implies (2\sqrt{xy})^2 \leq (x + y)^2 \implies \sqrt{xy} \leq \frac{x + y}{2},$$

as needed. We want to prove that

$$\sqrt{xy} = \frac{x+y}{2} \iff x = y.$$

Thus, we have

$$\sqrt{xy} = \frac{x+y}{2} \iff 4xy = x^2 + 2xy + y^2 \iff (x-y)^2 = 0 \iff x = y,$$

as needed. $\square$

Exercise 6. Let A and B be two nonempty bounded sets of real numbers. Let $C := \{a+b : a \in A, b \in B\}$. Show that C is a bounded set and that

$$\sup C = \sup A + \sup B \ \wedge \ \inf C = \inf A + \inf B.$$

Proof. Let A and B be any nonempty bounded subsets of $\mathbb{R}$. Define

$$C := \{a + b : a \in A \text{ and } b \in B\} \subset \mathbb{R}.$$

We want to show that C is a bounded subset. By the Least Upper Bound Property of $\mathbb{R}$, $\forall a \in A : \inf A \leq a \leq \sup A$ and $\forall b \in B : \inf B \leq b \leq \sup B$. Then

$$\forall a \in A \ \forall b \in B : \inf A + \inf B \leq a + b \leq \sup A + \sup B.$$

In other words,

$$\forall c \in C : \inf A + \inf B \leq c \leq \sup A + \sup B,$$

which is desired. Thus, C is a bounded subset of $\mathbb{R}$, as needed. We want to show that $\inf C = \inf A + \inf B$ and $\sup C = \sup A + \sup B$. By the Least Upper Bound Property of $\mathbb{R}$, $\forall c \in C : \inf C \leq c \leq \sup C$. Since $\inf A + \inf B$ is a lower bound for C, $\inf C \geq \inf A + \inf B$. Assume, for the purpose of contradiction, $\inf C > \inf A + \inf B$. Then

$$\exists a \in A : \inf C - \inf B > \inf A \implies \inf C - \inf B > a$$

and

$$\exists b \in B : \inf C - a > \inf B \implies \inf C - a > b.$$

In other words,

$$\exists a \in A \ \exists b \in B : \inf C > a + b \implies \exists c \in C : \inf C > c,$$

which is a contradiction. Thus, $\inf C = \inf A + \inf B$, as needed. Since $\sup A + \sup B$ is an upper bound for C, $\sup C \leq \sup A + \sup B$. Assume, for the purpose of contradiction, $\sup C < \sup A + \sup B$. Then

$$\exists a \in A : \sup C - \sup B < \sup A \implies \sup C - \sup B < a$$

and

$$\exists b \in B : \sup C - a < \sup B \implies \sup C - a < b.$$

In other words,

$$\exists a \in A \ \exists b \in B : \sup C < a + b \implies \exists c \in C : \sup C < c,$$

which is a contradiction. Thus, $\sup C = \sup A + \sup B$, as needed. $\square$

Exercise 7. Let
$$E = \{x \in \mathbb{R} : x > 0 \text{ and } x^3 < 2\}.$$

(a) Prove that E is bounded above.

(b) Let $r = \sup E$ (which exists by (a)). Prove that $r > 0$ and $r^3 = 2$. *Hint:* Adapt the proof used in Example 1.2.3.

Proof. Let $E := \{x \in \mathbb{R} : x > 0 \text{ and } x^3 < 2\}$. If $x = 1$, then $x > 0$ and $x^3 = 1 < 2$; hence, $x \in E$ ($E \neq \emptyset$).

(a) We want to prove that E is bounded above. Assume $x \in E$. Then
$$x^3 < 2 \implies x^3 < 8 \implies 8 - x^3 > 0 \implies (2-x)(4+2x+x^2) > 0.$$
That is, $4 + 2x + x^2 > 0$ and $2 - x > 0$ xor $4 + 2x + x^2 < 0$ and $2 - x < 0$. Given $x > 0$, $4 + 2x + x^2 > 0$ and hence $2 - x > 0$. Then
$$x < 2 \implies x \leq 2.$$
Thus, E is bounded above, as needed.

(b) Suppose $r = \sup E \in \mathbb{R}$. We want to prove that $r > 0$ and $r^3 = 2$. Given $\forall x \in E : x > 0$, $r = \sup E > 0$. Assume, for the purpose of contradiction, $r^3 < 2$. We want to find $h > 0$ such that $r + h > 0$ and $(r+h)^3 < 2$. Define $h \in \mathbb{R}$ such that
$$0 < h = \min\left\{\frac{1}{2}, \frac{2 - r^3}{3r^2 + 3r + 1}\right\} < 1.$$
We compute $h > h^2 > h^3 > 0$. Then $r + h > \max(r, h) > 0$ and
$$\begin{aligned}
(r+h)^3 &= r^3 + 3r^2h + 3rh^2 + h^3 \\
&< r^3 + 3r^2h + 3rh + h \\
&= r^3 + (3r^2 + 3r + 1)h \\
&\leq r^3 + (3r^2 + 3r + 1)\left(\frac{2 - r^3}{3r^2 + 3r + 1}\right) \\
&= r^3 + 2 - r^3 = 2.
\end{aligned}$$
Given $r + h > 0$ and $(r+h)^3 < 2$, $r + h \in E$. Then $r \leq \max(r, h) < r + h \in E$; hence, r is not an upper bound for E ($r \neq \sup E$), which is a contradiction. Assume, for the purpose of contradiction, $r^3 > 2$. We want to find $k > 0$ such that $r - k$ is an upper bound for E. Define $k \in \mathbb{R}$ such that
$$k = \frac{r^3 - 2}{3r^2} > 0.$$
We compute
$$r - k = \frac{3r^3 - r^3 + 2}{3r^2} = \frac{2r^3 + 2}{3r^3} > 0 \implies r > k.$$
Then
$$\begin{aligned}
(r-k)^3 &= r^3 - 3r^2k + 3rk^2 - k^3 \\
&> r^3 - 3r^2k + 2k^3 \\
&> r^3 - 3r^2k = r^3 - 3r^2\left(\frac{r^3 - 2}{3r^2}\right) \\
&= r^3 + 2 - r^3 = 2;
\end{aligned}$$

hence,

$$x > r - k \implies x^3 > (r-k)^3 > 2 \implies x \notin E.$$

Given the Principle of Contraposition,

$$x \in E \implies x^3 \leq (r-k)^3 \leq 2 \implies x \leq r - k < r.$$

Then $r - k$ is an upper bound for E; hence, r is not the least upper bound for E ($r \neq \sup E$), which is a contradiction. Thus, $r > 0$ and $r^3 = 2$, as needed.

$\square$